

9.5 Tree diagrams

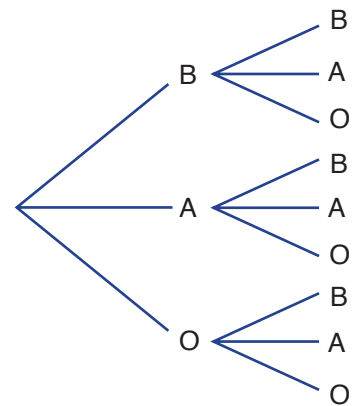
When multistage experiments consist of two or more components, a tree diagram can be used to list the sample space. While tables are often used for two-step experiments, a tree diagram can be extended for experiments with any number of steps.



Let's start: What's the difference?

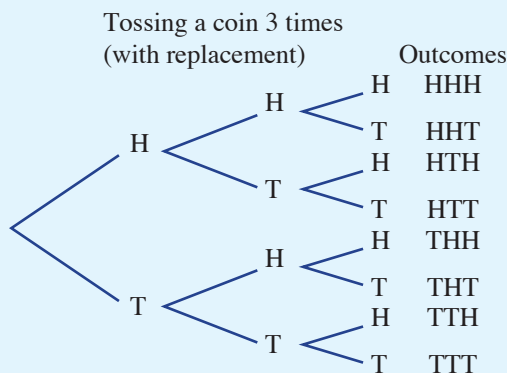
You are offered a choice of two pieces of fruit from a banana, an apple and an orange. You choose two at random. This tree diagram shows selection with replacement.

- How many outcomes will there be?
- How many of the outcomes contain two round fruits?
- How would the tree diagram change if the selection was completed without replacement? Would there be any difference in the answers to the above two questions? Discuss.

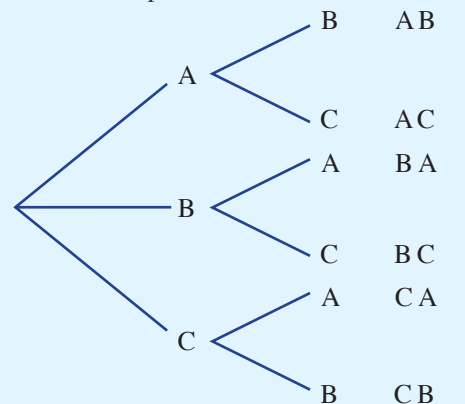


- **Tree diagrams** are used to list the sample space for multistage experiments with two or more steps.
- The outcomes for each stage of the experiment are listed vertically and each stage is connected with branches.

For example:



Selecting 2 letters from {A, B, C}
without replacement



In these examples, each set of branches produce outcomes which are all equally likely.

b i $\Pr(\text{Annabel and Brodie}) = \frac{2}{6}$
 $= \frac{1}{3}$

2 of the 6 outcomes contain Annabel and Brodie (A, B) and (B, A).

ii $\Pr(\text{Chris}) = \frac{4}{6}$
 $= \frac{2}{3}$

4 out of the 6 outcomes contain Chris.

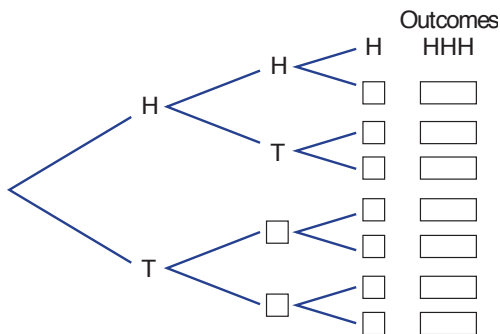
iii $\Pr(\text{Chris or Brodie}) = \frac{6}{6}$
 $= 1$

All of the outcomes contain at least one of Chris or Brodie.

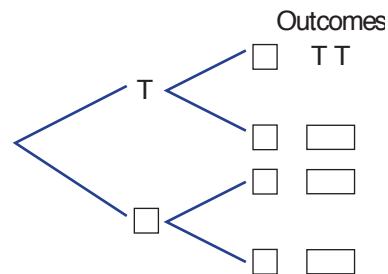
Exercise 9E

- 1** Find the total number of outcomes from these experiments *with replacement*. First, complete each tree diagram.

a Tossing a coin 3 times

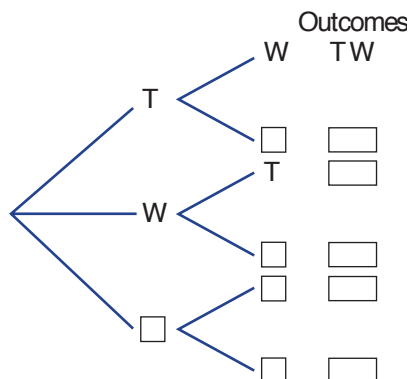


b Selecting 2 letters from the word TO

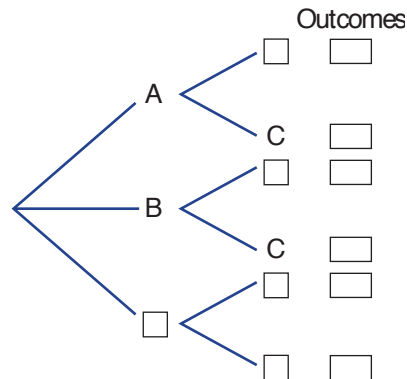


- 2** By first completing each tree diagram, find the total number of outcomes from these experiments *without replacement*.

a Selecting two letters from the word TWO

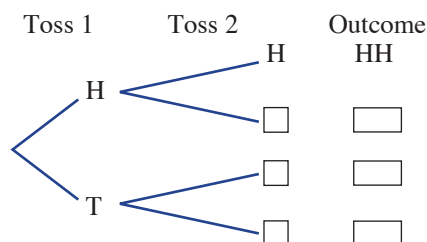


b Selecting two people from a group of three (A, B and C).



Example 8

- 3 A coin is tossed twice.
- Complete this tree diagram to show all the possible outcomes.
 - What is the total number of outcomes?
 - Find the probability of obtaining:
 - two heads
 - one head
 - at least one head
 - at least one tail.



Fluency

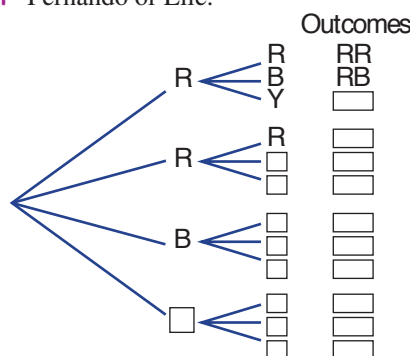
- 4 A spinner with three numbers, 1, 2 and 3, is spun twice.
- List the set of possible outcomes, using a tree diagram.
 - What is the total number of possible outcomes?
 - Find the probability of spinning:
 - two 3s
 - at least one 3
 - no more than one 2
 - two odd numbers.

Example 9

- 5 Two people are selected without replacement from a group of three: Donna (D), Elle (E) and Fernando (F).
- List all the possible combinations for the selection using a tree diagram.
 - Find the probability that the selection will contain:
 - Donna and Elle
 - Fernando
 - Fernando or Elle.

- 6 A drawer contains 2 red socks (R), 1 blue sock (B) and 1 yellow sock (Y) and two socks are selected at random without replacement.

- Complete this tree diagram.
- Find the probability of obtaining:
 - a red sock and a blue sock
 - two red socks
 - any pair of socks of the same colour
 - any pair of socks of different colour.

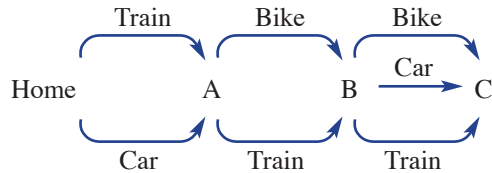


- 7 A student who has not studied for a multiple-choice test decides to guess the answers for every question. There are three questions, and three choices of answer (A, B and C) for each question. If only one of the possible choices (A, B or C) is correct for each question, find the probability that the student guesses:
- 1 correct answer
 - 2 correct answers
 - 3 correct answers
 - 0 correct answers

- 8 A discount supermarket shelf contains a large number of tomato tins and peach tins with no labels. There are an equal number of tins all mixed together on the same shelf. You select four tins in a hurry. Use a tree diagram to help find the probability of selecting the correct number of tins of tomatoes and/or peaches for each of these recipe requirements.
- You need four tins of tomatoes for a stew.
 - You need four tins of peaches for a peach crumble.
 - You need at least three tins of tomatoes for a bolognese.
 - You need at least two tins of peaches for a fruit salad.
 - You need at least one tin of tomatoes for a vegetable soup.

Problem-solving

- 9 Michael needs to deliver parcels to three places (A, B and C in order) in the city. This diagram shows the different ways that he can travel.



Problem-solving

- a Draw a tree diagram showing all the possible combinations of transportation.
- b What is the total number of possible outcomes?
- c Find the probability that Michael will use:
 - i the train all three times
 - ii the train twice
 - iii his bike once
 - iv different transport each time
 - v a car at least once.



- 10 If a coin is tossed four times, use a tree diagram to find the probability that you receive:
 - a 0 tails
 - b 1 tail
 - c 2 tails
 - d 3 tails
 - e 4 tails
- 11
 - a A coin is tossed 5 times. How many outcomes will there be?
 - b A coin is tossed n times. Write a rule for the number of outcomes in the sample space.
- 12 Use a tree diagram to investigate the probability of selecting two counters from a bag of 3 black and 2 white counters if the selection is drawn
 - a with replacement
 - b without replacement
 Is there any difference?

Reasoning

Enrichment: Selecting matching clothes

- 13 A man randomly selects a tie from his collection of one green and two red ties, a shirt from a collection of one red and two white, and either a red or black hat. Use a tree diagram to help find the probability that the man selects a tie, shirt and hat according to the following descriptions:
 - a a red tie, red shirt and black hat
 - b all three items red
 - c one item red
 - d two items red
 - e at least two items red
 - f green hat
 - g green tie and a black hat
 - h green tie or a black hat
 - i not a red item
 - j red tie or white shirt or black hat

